

terials scientists to use previously unexplored properties of these films. The new discovery could greatly improve the performance of certain types of integrated circuits.

By using a new technique that altered the deposition process of diamond films, researchers were able to reduce the temperature to close to 400°C and also tune the thermal properties of diamond films by controlling their grain size. This permitted the eventual combination of diamond with two other important materials, graphene and gallium nitride.

In the other study, the researchers used the same technology to combine diamond thin films with gallium nitride, which is used extensively in high-power LEDs. After depositing a 300nm thick diamond film on a gallium nitride substrate, they noticed a considerable improvement in the thermal performance. The result is remarkable because a temperature difference of just a few degrees within an integrated circuit can cause a noticeable change in performance.

The common link between these experiments is that the researchers found new ways of dissipating heat more effectively while using less energy.

Solar power from microscopic silver

According to a study at the University of California, a new type of solar cell can possibly be developed from microscopic fractal trees grown from silver. Fractals are patterns that repeat over multiple lengths. In this case, microscopic silver with a branching pattern that grows more branches is used. According to researchers, these structures can make more efficient solar cells.

The silver trees are coated with light-absorbing polymers. When photons hit the polymer coat, they produce short-lived electrons and holes in the polymer. The positively charged holes are collected through the silver branches, while the electrons move to the opposite electrode to create an electrical potential. The nanosized silver trees have a large surface area.

The research lab will fabricate the cells, which will be characterised by collaborators from two other universities. It will also carry out computer modelling on the systems.

Highly efficient solar collectors

An Arizona company claims that highly efficient solar collectors designed by it could cut solar farm costs by \$1 per watt of generation capacity by 2020. The design uses a framework of mirrors to focus sunlight onto small areas of expensive multilayer solar cells. The company is now building a 20kW prototype with a lightweight frame to track the sun on a test site in the Arizona desert.

The mirrors used in the design are made out of thin sheets of cheap float glass. Heating softens the sheets so that they slump into a parabolic mold. Once cooled, these are coated with a reflective metal. The prototype uses a 3.1m wide mirror that focuses the sunlight onto a spherical glass receiver. The receiver distributes the sunlight to an array of high-efficiency, three-layer solar cells.

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